Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cubes, cuboids, pyramids and cones. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	4
Topic	Three Dimensional Shapes
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones
Theme	—
Curriculum objectives	• Identify the properties of cube and cuboids • Identify the properties of pyramids and cones
Previous knowledge	• Students have been taught the differences between cube and cuboid.
Lesson sequence	• Identify the properties of cube and cuboids — the teacher writes the properties of cubes and cuboids on the board; the student take notes in their notebooks • Identify the properties of pyramids and cones — the teacher draws the pyramids and the cone on the board and writes their properties; the student take them down, as in Step II
Instructional materials	• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.
Evaluation	• How many vertices, has a cuboid? • How many edges has a cube? • Draw a cube and cuboid then label. • Draw an hexagonal base pyramid and list the properties. • How many vertices has a tetrahedron.

Lesson objectives

1.	Identify the properties of cube and cuboids
2.	Identify the properties of pyramids and cones

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cubes, cuboids, pyramids and cones

Class: JSS1 **Subject:** Mathematics **Week:** 4

Section 1: Cubes and Cuboids: Faces, Edges and Vertices

Around you are solid shapes that take up space. A box, a dice, a brick and a matchbox are all three-dimensional shapes, which means they have length, width and height. In this section you will look closely at two of them: the cube and the cuboid. To describe either one, you need three key ideas.

• A face is one of the flat surfaces of the shape.
• An edge is where two faces meet — the straight line between them.
• A vertex (plural: vertices) is a corner, where edges meet.

Let us see how these appear on a cube. A cube is the shape of a dice. Every one of its faces is a square, and all its faces are the same size. Count them: a cube has 6 faces. Where the faces meet, there are straight edges, and a cube has 12 edges — all of them the same length. Where the edges meet, there are corners, and a cube has 8 vertices.

Now picture a cuboid, the shape of a brick or a box of cereal. A cuboid also has flat faces, but they are rectangles, and not all of them are the same size: the front and back are identical, the two sides are identical, and the top and bottom are identical. Like a cube, a cuboid has 6 faces, 12 edges and 8 vertices. The difference is that the edges of a cuboid are not all equal — there are three different lengths (length, width and height), and each length is repeated four times.

So a cube and a cuboid are actually close cousins: both have 6 faces, 12 edges and 8 vertices. What separates them is that every face of a cube is an equal square, while a cuboid has rectangular faces that may differ in size.

Worked example 1
A dice is a cube. State the number of faces, edges and vertices on it.

Solution
A cube has 6 square faces. It has 12 edges, all of equal length. It has 8 vertices (corners). So the dice has 6 faces, 12 edges and 8 vertices.

Worked example 2
A cereal box is a cuboid. It has length 30 cm, width 10 cm and height 20 cm.
(a) How many faces does it have, and what shapes are they?
(b) How many edges of length 30 cm does it have?

Solution
(a) A cuboid has 6 faces, and they are rectangles.
(b) The length 30 cm appears four times, on the four edges that run along the length of the box. So there are 4 edges of length 30 cm.

Practice

1. Write down the number of faces, edges and vertices of a cube.
2. A brick is a cuboid. (a) How many faces does it have and what shape is each face? (b) A cuboid has edges of three different lengths — length, width and height. How many edges of each length does it have?
3. A cube and a cuboid both have the same number of faces, edges and vertices. State these three numbers, and explain the one way in which a cube is different from a cuboid.

Section 2: Pyramids and Cones: Faces, Edges and Vertices

A pyramid is a three-dimensional shape with one flat base and triangular side faces that all meet at a single point called the apex. The name of a pyramid tells you the shape of its base. For example, a pyramid with a triangular base is a triangular pyramid, and one with a square base is a square pyramid.

Every pyramid is described by three properties:
- Faces: the flat surfaces. A square pyramid has a square base plus four triangular side faces, so 5 faces in total.
- Vertices: the corners where faces meet. A square pyramid has 5 vertices — the 4 corners of the base and the apex.

- Edges: the straight lines where two faces meet. A square pyramid has 4 edges around the base and 4 edges slanting up to the apex, so 8 edges in total.

No matter what shape the base is, a pyramid always has just one apex, and its side faces are always triangles.

A cone is like a pyramid whose base is a circle instead of a polygon. A cone has:
- Faces: a cone has one flat circular base and one curved surface. The curved surface is not a flat face, so a cone is described as having 1 face.
- Vertices: a cone has one vertex, the pointed apex at the top.
- Edges: a cone has one edge, the circle where the curved surface meets the flat base.

The two key things to remember about a cone are that its base is a circle and its one curved surface slides up to a single point. Because the base is curved, a cone has only one edge (the circle), unlike a pyramid which has several straight edges.

Example: Count the faces, edges and vertices of a triangular pyramid.

Step 1 — Identify the base. The base is a triangle, so the pyramid has 1 triangular base.
Step 2 — Count the side faces. There is one triangular side face for each side of the base, so 3 side faces.
Step 3 — Add the faces together. 1 base + 3 side faces = 4 faces in total.
Step 4 — Count the vertices. The base triangle has 3 corners, and there is 1 apex on top, so 3 + 1 = 4 vertices.
Step 5 — Count the edges. The base has 3 edges and each corner of the base connects to the apex, giving 3 more edges. So 3 + 3 = 6 edges.

So a triangular pyramid has 4 faces, 4 vertices and 6 edges.

Example: Describe the faces, edges and vertices of a cone.

Step 1 — Look at the base. The base of a cone is a circle, so the cone has 1 flat face.
Step 2 — Look at the rest of the shape. The curved surface wraps around from the base up to the point, so that is 1 more surface.
Step 3 — Count the edges. The only edge is the circle where the curved surface meets the base, so the cone has 1 edge.
Step 4 — Count the vertices. The cone has one point at the top, so it has 1 vertex.

So a cone has 1 face, 1 edge and 1 vertex.

Practice

1. A pyramid has a square base. How many faces, vertices and edges does it have in total? Write your answer as three numbers, e.g. "x faces, y vertices, z edges".
2. Explain in one sentence what shape the side faces of any pyramid are, and what shape the base of any cone is.
3. A cone is sometimes said to have "one face, one edge and one vertex." In your own words, identify each of these three parts of a cone.

Check your work

Attempt every question before you read this page.

Cubes and Cuboids: Faces, Edges and Vertices

1. A cube has 6 faces, 12 edges and 8 vertices.
2. (a) 6 faces, each a rectangle. (b) 4 edges of each length, so 4 length-edges, 4 width-edges and 4 height-edges.

   A cuboid has 6 faces, all rectangles. Its edges come in three lengths (length, width, height). There are 12 edges in total, so 12 ÷ 3 = 4 edges of each length.
3. Both have 6 faces, 12 edges and 8 vertices. The difference is that every face of a cube is an equal square, while the faces of a cuboid are rectangles that may differ in size (and the edges of a cube are all equal, while a cuboid has three different edge lengths).

Count on the shapes: each has 6 flat faces, 12 edges and 8 vertices. In a cube all six faces are identical squares and all edges are equal. In a cuboid the faces are rectangles and not all edges are equal — there are three lengths, each repeated four times.

Pyramids and Cones: Faces, Edges and Vertices

1. 5 faces, 5 vertices, 8 edges

   A square pyramid has 1 square base and 4 triangular side faces, so $1 + 4 = 5$ faces. It has 4 base corners plus 1 apex, so $4 + 1 = 5$ vertices. It has 4 edges around the base and 4 slanting edges to the apex, so $4 + 4 = 8$ edges.
2. The side faces of any pyramid are triangles, and the base of any cone is a circle.
3. A cone has 1 face, 1 edge and 1 vertex.

   The one face is the flat circular base (the curved surface is not counted as a flat face). The one edge is the circle where the curved surface meets the base. The one vertex is the single pointed apex at the top of the cone.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-7efef142	**Date**	
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____